

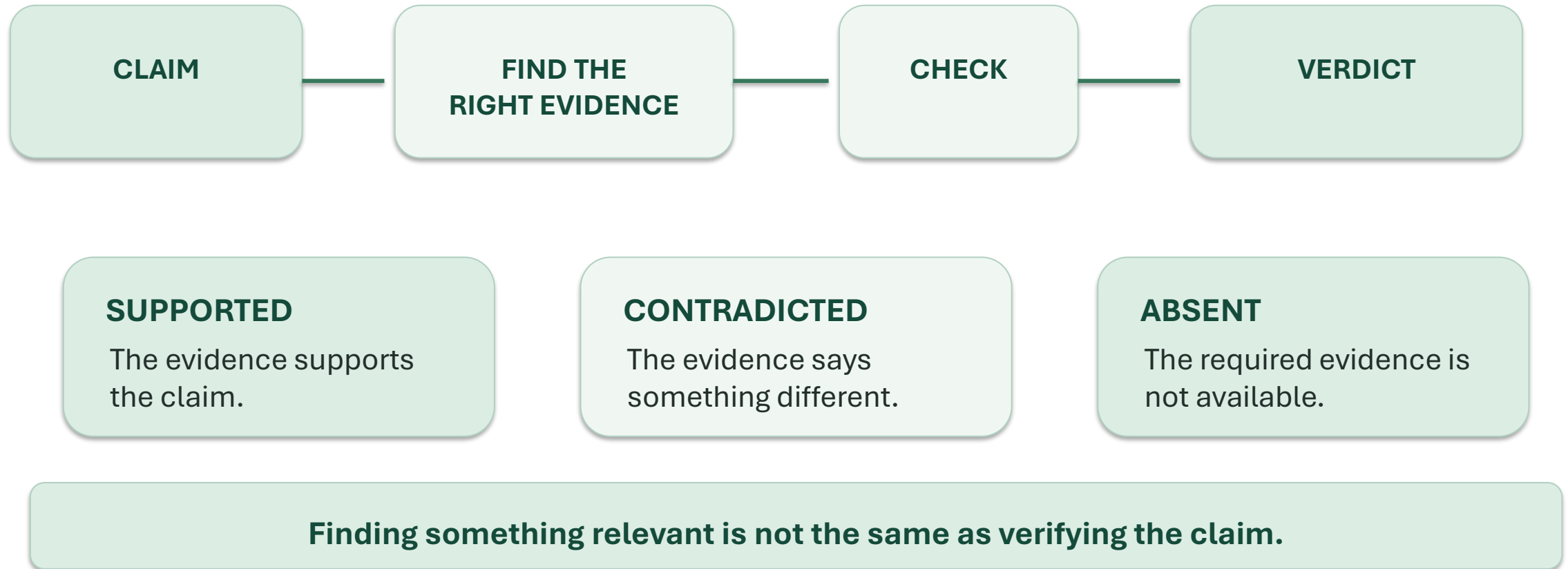
CHAPTER 7

Claim Verification and Calibrated Abstention



What Is Claim Verification?

Check a claim against the evidence that can actually answer it.



First, Make Sure We Are Checking the Right Thing

A correct number from the wrong model, metric or sample is still the wrong answer.

QUESTION

“What was the selected region’s Brier contrast on confirmation?”

WHAT MUST MATCH

Region

Metric

Confirmation sample

Model / result version

THEN RETRIEVE

Once those pieces are clear, query the KG for the exact record and its source.

Verification starts by identifying exactly what the question refers to.

Four Ways to Verify

Use the simplest method that can answer the question.

1 LOOK UP	The answer already exists.	KG, database, source record	What was the confirmed Brier loss?
2 COMPUTE	The answer can be calculated.	Arithmetic, SQL, statistical test	How much worse is this region?
3 PREDICT / SIMULATE	The answer is not directly observed.	Statistical / ML model, simulation	What happens under a future stress?
4 SEMANTIC JUDGMENT	The question is genuinely about meaning.	Validated semantic verifier	Does this paraphrase preserve the limitation?

Don't use an LLM to judge something you can look up, calculate or measure.

Semantic Verification Is the Last Resort

Sometimes meaning cannot be reduced to an exact lookup or calculation.

WHEN TO USE IT

Paraphrases

Meaning equivalence

Claim–evidence matching
that cannot be resolved
deterministically

HOW TO VALIDATE IT

Test on labeled held-out
cases

Measure false accepts and
false rejects

Calibrate the decision
threshold

WHEN UNSURE

Abstain or send the case for
review.

A semantic score is not
evidence by itself.

A semantic verifier is still a model. It has to earn our trust.

The KG Finds the Record; the Source Supports the Answer

Keep the answer tied to the evidence it came from.



EXACT VALUES

Keep the metric, population, split and model version with the number.

PROVENANCE

Keep the source location so a reviewer can inspect the original evidence.

The graph is the index. The original source remains the evidence.

When Exact Retrieval Is Not Enough, Geometry Can Help

Use geometry to find or compare candidates — then check them against source-backed evidence.

1. FIND CANDIDATES

A wording or relationship may not match the stored graph exactly.

Geometry can rank plausible entities, relations or paths.

2. CHECK THE PATH

For a multi-step claim, inspect whether the relationships fit together and whether scope or versions change along the path.

3. RETURN TO EVIDENCE

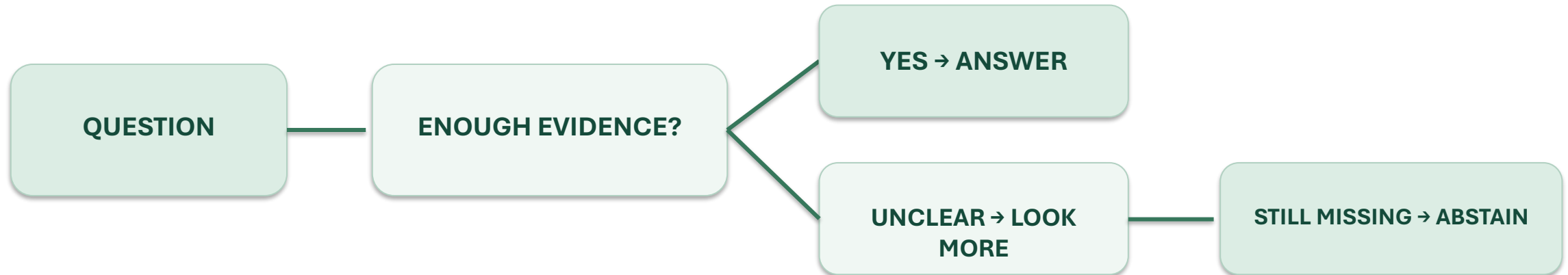
A geometric score can point us somewhere useful.

The source-backed record determines what we can actually claim.

Geometry helps us search and diagnose. It does not replace evidence.

Answer, Look Further or Abstain

Verification does not require an answer when the evidence is not there.



IF EVIDENCE IS AMBIGUOUS

Retrieve more evidence or run the deterministic test that is missing.

IF EVIDENCE IS ABSENT

Say what is missing instead of guessing.

A reliable verifier knows when it does not have enough evidence.